

Introduction to R and RStudio

Computing Fundamentals for Social Sciences

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Outline

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What is R?

- R is a free software environment for statistical computing and graphics.
- Widely used in data science, statistics, and research.
- Large ecosystem of packages (CRAN).
- Community-driven, open source.

Installing R and RStudio

- Download R: <https://cran.r-project.org>
- Download RStudio:
<https://posit.co/download/rstudio-desktop/>
- R: the language and computation engine.
- RStudio: an IDE for R (easier coding, plotting, and project management).

RStudio Interface

- Console: where code runs.
- Script editor: write and save code.
- Environment: view variables and data.
- Plots/Files/Help: visualization and resources.

Basic Operations

Variables

```
x <- 5
```

```
y <- 3
```

```
sum_xy <- x + y
```

Vectors

```
numbers <- c(1, 2, 3, 4, 5)
```

```
numbers * 2
```

Functions

```
mean(numbers)
```

```
sd(numbers)
```

What are R Packages?

- R packages are collections of functions, data, and documentation.
- Extend R's capabilities (e.g., data visualization, statistical models).
- Most packages are hosted on CRAN (Comprehensive R Archive Network).
- Popular packages: `dplyr`, `ggplot2`, `tidyr`, `shiny`.

Installing and Loading Packages

```
# Install a package (from CRAN)
install.packages("ggplot2")

# Load a package
library(ggplot2)

# Example: scatter plot from iris dataset
ggplot(iris, aes(x = Sepal.Length, y = Sepal.Width)) +
  geom_point()
```


Multiple Packages

```
# Install multiple packages at once
install.packages(c("dplyr", "tidyr"))

# Load them
library(dplyr)
library(tidyr)

# Example with dplyr
iris %>% group_by(Species) %>%
  summarise(avg_length = mean(Sepal.Length))
```

Data Types in R

```
# Numeric
a <- 42

# Character
b <- "Hello"

# Logical
c <- TRUE

# Factor (categorical)
colors <- factor(c("red", "blue", "red"))

# Checking type
class(colors)
levels(colors)
```

Data Frames

```
# Creating a small dataset
df <- data.frame(
  name = c("Alice", "Bob", "Charlie"),
  age = c(25, 30, 22),
  score = c(90, 85, 88)
)
df

# Accessing columns
df$name
df$age
```

Built-in Datasets

```
# Load iris dataset (comes with R)
data(iris)
head(iris)
summary(iris)

# Load mtcars dataset
data(mtcars)
head(mtcars)
```

Importing Data

```
# Simulated CSV in R
my_data <- data.frame(
  id = 1:5,
  name = c("A", "B", "C", "D", "E"),
  value = c(10, 20, 15, 25, 30)
)

# Save to CSV
write.csv(my_data, "example.csv", row.names = FALSE)

# Read CSV
data_from_file <- read.csv("example.csv")
head(data_from_file)
```

dplyr Basics

```
library(dplyr)

# Filter rows
iris %>% filter(Species == "setosa")

# Select columns
iris %>% select(Sepal.Length, Species)

# Mutate (add new column)
iris %>% mutate(Sepal.Area = Sepal.Length * Sepal.Width)

# Group and summarise
iris %>% group_by(Species) %>%
  summarise(mean_length = mean(Sepal.Length))
```

Basic Plots

```
# Scatter plot (iris dataset)
plot(iris$Sepal.Length, iris$Sepal.Width)

# Histogram
hist(iris$Sepal.Length)
```

ggplot2 Example

```
library(ggplot2)

# Scatter plot with colors
ggplot(iris, aes(x = Sepal.Length, y = Sepal.Width, color =
  Species)) +
  geom_point()

# Histogram
ggplot(iris, aes(x = Sepal.Length)) +
  geom_histogram(bins = 20, fill = "skyblue")
```


Saving Work in R

```
# Save R objects
save(iris, file = "iris.RData")

# Load them later
load("iris.RData")

# Save script in RStudio: File -> Save
# Export plot: in Plots panel -> Export
```

Exercises (Guided)

- 1 Load the `mtcars` dataset and calculate the average mpg.
- 2 Create a histogram of `Sepal.Length` in `iris`.
- 3 Filter `iris` to only show `versicolor`.
- 4 Save a data frame to CSV and reload it.
- 5 Using `dplyr`, calculate the mean `Petal.Length` for each species.

Solutions

```
# 1. Average mpg
mean(mtcars$mpg)

# 2. Histogram of Sepal.Length
hist(iris$Sepal.Length)

# 3. Filter versicolor
subset(iris, Species == "versicolor")

# 4. Save & reload
write.csv(iris, "iris_copy.csv", row.names = FALSE)
iris2 <- read.csv("iris_copy.csv")

# 5. Mean Petal.Length per species
library(dplyr)
iris %>% group_by(Species) %>%
  summarise(mean_petal = mean(Petal.Length))
```